ENUNCIADO DEL EXAMEN

Numerical Analysis – Preliminary Exam

Department of Mathematical Sciences

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Instructions: Answer all four of the following problems. This is a closed

book, closed notes exam. Show all of your work for full credit. No calculators

or computers are allowed.

1. (a) Consider the fixed point iteration scheme used to solve f (x) = 0 (i.e. find

α such that f (α) = 0)

xn+1 = g(xn) = xn − f (xn)/c,

where c is a constant value. Under what conditions on the value of c will this

scheme be locally convergent? What will the convergence rate be in general?

Is there any value of c for which the scheme will converge quadratically? If

so, show details.

(b) Show that Newton's method for solving f (x) = 0 corresponds to a special

case of fixed point iteration (i.e. identify the corresponding function g(x)

where xn+1 = g(xn)). Show that if f ′(α) 6 = 0 the convergence is locally

quadratic.

(c) Solve x2 = 0 with Newton's Method (start with the guess x0 = 1 and determine an explicit formula for the kth iterate) and discuss the convergence

properties in this case.

2. Derive the maximum time step allowed for numerical stability of Euler's

method applied to the initial value problem stated below.

dy

dt = −5y,

y(0) = 1,

Is there a numerical stability requirement if the Backward Euler method were

used? Explain why or why not.

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3. (a) Suppose that you have at your disposal your favorite algorithm to solve

a nonsingular linear system of equations. Describe a numerical procedure

based on the finite difference method for solving the following boundary

value problem

$u'' + u = 0$,

$u(0) = 0$,

$u(1) = 1$.

that could make use of your linear solver (ignore for now that it has an

analytical solution). Use a uniform grid with $N + 1$ equally-spaced points

with $x_j = (j - 1)/N$ for $j = 1, \ldots, N + 1$.

(b) Now suppose that instead of a linear solver you have at your disposal your favorite initial value problem solver (that solves a system of first order differential equations, e.g. Euler's method, Runge-Kutta, etc.) and your favorite nonlinear solver (e.g. bisection, Newton's method, secant method, ...). Outline the details of a procedure that you could use in this case to solve the above boundary value problem. Note that original problem involves a second order differential equation so be sure to provide details on the appropriate conversion required. Also, be sure to describe how you will obtain all the necessary ingredients for the nonlinear solver you choose (e.g. function values, derivatives, Jacobian, ...)

(c) Finally, suppose that you wanted to solve the following eigenvalue problem numerically (ignore for now that it has a relatively easy to find analytical solution) to obtain the three smallest positive eigenvalues λ and corresponding eigenfunctions u

$u'' + \lambda u = 0,$

$u(0) = 0,$

$u(1) = 0.$

If you had at your disposal your favorite eigenvalue solver (i.e. a numerical algorithm that, given a matrix A, finds the eigenvalues λ and eigenvectors x

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of $Ax = \lambda x$) describe how you could make use of this to solve the differential eigenvalue problem. If you did not have an eigenvalue solver but had any of

your other favorite algorithms listed above describe how you could solve the

differential eigenvalue problem (again ignoring an analytical solution).

4. Consider the linear least squares problem where you wish to find x such that

||b − Ax||2

2,

is minimized. Here A is a full rank m × n matrix (m > n), b is a given

vector of length m and x is a solution vector of length n.

(a) Suppose A has the QR factorization A = QR where Q is an m × m

orthogonal matrix and R is an m × n matrix with the upper n × n portion

of R upper triangular and the remaining portion of R zero. Show how the

solution of the least squares problem can be found by solving an n × n

triangular system that makes use of Q and/or R.

(b) Consider the QR factorization of the matrix given below

A =

1 10

1 15

1 20

= QR =

1 √3 − 1 √2

1 √6

1 √3 0 − 2 √6

1 √3

1 √2

1 √6

√3 15√3

0 5√2

0 0

.

Describe in words, generally but not doing specific computations, two ways

in which the QR factorization of a matrix could be obtained.

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SOLUCIÓN DETALLADA

Solución — GMU Numerical Analysis Preliminary Exam, Agosto 2011

Formato: responder los 4 problemas (todos obligatorios), libro/notas cerrados, sin calculadora ni computador.

Sección principal del temario: ecuaciones no lineales (punto fijo / Newton), EDO (estabilidad de Euler explícito/implícito), BVP (diferencias finitas, shooting, problema de valores propios) y álgebra lineal numérica (mínimos cuadrados vía QR).

Problema 1 — Iteración de punto fijo y método de Newton

(a) $g(x)=x-f(x)/c$. Convergencia local requiere $|g'(\alpha)|<1$, con $g'(x)=1-f'(x)/c$, es decir $$\left|1-\frac{f'(\alpha)}{c}\right|<1 \iff 0<\frac{f'(\alpha)}{c}<2,$$ o sea c debe tener el mismo signo que $f'(\alpha)$ y satisfacer $|c|>|f'(\alpha)|/2$. En general la convergencia es **lineal** (orden 1), con factor asintótico $|g'(\alpha)|=|1-f'(\alpha)/c|$.

Convergencia cuadrática es posible si elegimos $c=f'(\alpha)$: entonces $g'(\alpha)=0$ y, por Taylor, $e_{n+1}\approx \tfrac12 g''(\alpha) e_n^2$ con $g''(x)=-f''(x)/c$, dando convergencia cuadrática (si $f''(\alpha)\neq0$). En la práctica α es desconocido, así que esto es una construcción teórica.

(b) El método de Newton usa $c_n=f'(x_n)$ (no constante), es decir $g(x)=x-f(x)/f'(x)$. Entonces $$g'(x)=1-\frac{f'(x)^2-f(x)f''(x)}{f'(x)^2}=\frac{f(x)f''(x)}{f'(x)^2}.$$ En $x=\alpha$, $f(\alpha)=0\Rightarrow g'(\alpha)=0$ (si $f'(\alpha)\neq0$), y por Taylor $e_{n+1}=\tfrac12 g''(\alpha)e_n^2+O(e_n^3)$: **convergencia cuadrática** (orden 2), suponiendo $f''(\alpha)\neq0$.

(c) $f(x)=x^2$, $f'(x)=2x$. Newton: $x_{n+1}=x_n-\dfrac{x_n^2}{2x_n}=\dfrac{x_n}{2}$. Con $x_0=1$: $\boxed{x_n=2^{-n}}$.

Aquí $x_n\to0$ pero con $e_{n+1}=\tfrac12 e_n$: **convergencia solo lineal**, no cuadrática, porque $\alpha=0$ es una **raíz doble** de f ($f'(0)=0$), violando la hipótesis $f'(\alpha)\neq0$ del inciso (b). Esto ilustra que Newton pierde el orden cuadrático en raíces múltiples.

Problema 2 — Estabilidad de Euler explícito e implícito

$y'=-5y$, $y(0)=1$.

Euler explícito: $y_{n+1}=y_n+h(-5y_n)=(1-5h)y_n$. Factor de amplificación $R=1-5h$. Estabilidad: $|1-5h|\le1\iff 0\le h\le \tfrac25$. $$\boxed{h_{\max}=2/5=0.4.}$$

Euler implícito (backward): $y_{n+1}=y_n-5h\,y_{n+1}\Rightarrow y_{n+1}=\dfrac{y_n}{1+5h}\(. \)R=\dfrac1{1+5h}\(. Para todo \)h>0\(, \)0no hay restricción de estabilidad (el método es incondicionalmente estable / A-estable para esta ecuación), aunque la precisión se degrada si \(h\) es grande.

Problema 3 — BVP: tres formulaciones numéricas

(a) Diferencias finitas + solver lineal. \(u''+u=0\), \(u(0)=0\), \(u(1)=1\), malla uniforme \(x_j=(j-1)/N\), \(h=1/N\). Diferencia central en nodos interiores \(j=2,\dots,N\): \[\frac{u_{j-1}-2u_j+u_{j+1}}{h^2}+u_j=0\ \Longrightarrow\ u_{j-1}+(h^2-2)u_j+u_{j+1}=0,\] con \(u_1=0\), \(u_{N+1}=1\) conocidos. Esto da un sistema tridiagonal \(A\mathbf{u}=\mathbf{b}\) de tamaño \((N-1)\), resoluble con el solver lineal disponible.

(b) IVP solver + solver no lineal (shooting). Reescribir como sistema de primer orden: \(y_1=u,\ y_2=u'\), \(y_1'=y_2,\ y_2'=-y_1\), con \(y_1(0)=0\) conocido y \(y_2(0)=s\) desconocido. Integrar con el IVP solver de \(x=0\) a \(x=1\) para obtener \(y_1(1;s)\). Definir \(F(s)=y_1(1;s)-1\) y usar el solver no lineal (bisección/Newton/secante) para hallar \(s^*\) con \(F(s^*)=0\). Para Newton en \(s\) se necesita \(F'(s)=\partial y_1(1;s)/\partial s\): al ser el problema **lineal**, las ecuaciones de sensibilidad \(\partial y/\partial s\) satisfacen el mismo sistema ODE, con condición inicial $(\partial y_1/\partial s,\partial y_2/\partial s)(0)=(0,1)$, y se integran junto con el sistema original (o, alternativamente, se aproxima \(F'(s)\) por diferencias finitas perturbando \(s\) y resolviendo el IVP de nuevo).

(c) Problema de valores propios. \(u''+\lambda u=0\), \(u(0)=u(1)=0\).

- *Con solver de valores propios:* discretizar igual que en (a) para obtener

\(\dfrac{u_{j-1}-2u_j+u_{j+1}}{h^2}=-\lambda u_j\), es decir un problema de valores propios matricial \(A\mathbf{u}=\lambda\mathbf{u}\) con \(A\) tridiagonal simétrica. Pasar \(A\) al solver de valores propios y tomar los tres \(\lambda\) positivos más pequeños (que aproximan a los del problema continuo, \(\lambda_n=(n\pi)^2\)).

- *Sin solver de valores propios (shooting):* fijar \(u(0)=0\), \(u'(0)=1\)

(normalización arbitraria), integrar con el IVP solver hasta \(x=1\) para un \(\lambda\) de prueba, y definir \(F(\lambda)=u(1;\lambda)\). Buscar raíces de \(F(\lambda)\) (bisección/Newton/secante) escaneando \(\lambda>0\) para detectar cambios de signo; cada raíz da un valor propio. Tomar las tres primeras raíces positivas encontradas.

Problema 4 — Mínimos cuadrados vía factorización QR

(a) Con \(A=QR\) (\(Q\) ortogonal \(m\times m\)), $\|b-Ax\|^2=\|Q^T(b-QRx)\|^2 =\|Q^Tb-Rx\|^2\((norma invariante bajo \)Q^T$ ortogonal). Escribiendo \

$Q^Tb=\begin{pmatrix}c_1\\c_2\end{pmatrix}$ ($c_1\in\mathbb R^n$, $c_2\in\mathbb R^{m-n}$) y $R=\begin{pmatrix}R_1\\0\end{pmatrix}$ (R_1 triangular superior $n\times n$): $$\|Q^Tb-Rx\|^2=\|c_1-R_1x\|^2+\|c_2\|^2.$$ Esto se minimiza (en x) cuando $c_1-R_1x=0$, es decir resolviendo el sistema triangular $R_1x=c_1$ por sustitución hacia atrás; el residuo mínimo es $\|c_2\|$ (independiente de x).

(b) Dos formas generales de obtener la factorización QR de una matriz:

1. **Proceso de Gram-Schmidt** (clásico o modificado) aplicado a las columnas

de A, ortonormalizándolas sucesivamente: Q se forma con los vectores ortonormales resultantes y R con los coeficientes de proyección.

1. **Reflexiones de Householder** (o rotaciones de Givens): se aplican

transformaciones ortogonales sucesivas que anulan las entradas bajo la diagonal, triangularizando A y acumulando las transformaciones para formar Q.